

Workshop 2: Regression models

TECH6

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October 22, 2026

See GitHub repository for notebooks and data:

<https://github.com/richardfoltyn/TECH6-H26>

Exercise 1: Predicting house prices with linear models

In this exercise, you will work with the Ames housing dataset, which we already encountered in the lectures. Your task is to evaluate the following three linear models in terms of their performance when predicting house prices:

1. Linear regression without any regularization
2. Ridge regression
3. Lasso

Data description

The data is stored in the file `data/ames_houses.csv` and can be loaded as follows:

```
[1]: import pandas as pd

# Use this path to use the CSV file from the data/ directory
file = '../data/ames_houses.csv'

df = pd.read_csv(file, sep=',')

# Variables used in the analysis
variables = [
    'LotArea',
    'LivingArea',
    'Bathrooms',
    'Bedrooms',
    'SalePrice',
    'OverallQuality',
    'BuildingType',
    'YearBuilt',
    'CentralAir',
]

# Drop rows with any missing observation
df = df.dropna(subset=variables)

# Drop observations with large living or lot area
df = df.query('LivingArea <= 350 & LotArea <= 5000')
```

```
print(f'Number of observations: {df.shape[0]:,d}')
```

Number of observations: 2,755

The included variables are a simplified subset of the original data (see [here](#) for a detailed description of the original variables):

1. LotArea: Lot size in square meters
2. Neighborhood: Physical locations within Ames city limits
3. OverallQuality: Rates the overall material and finish of the house (1 = very poor, 10 = excellent)
4. OverallCondition: Rates the overall condition of the house (1 = very poor, 10 = excellent)
5. YearBuilt: Original construction date
6. YearRemodeled: Remodel date (same as construction date if no remodeling or additions)
7. BuildingType: Type of dwelling
8. CentralAir: Central air conditioning (string, Y/N)
9. LivingArea: Above-grade (ground) living area in square meters
10. Bathrooms: Full bathrooms above grade
11. Bedrooms: Bedrooms above grade (does not include basement bedrooms)
12. Fireplaces: Number of fireplaces
13. SalePrice: Sale price in thousands of USD
14. YearSold: Year sold
15. MonthSold: Month sold
16. HasGarage: Flag indicating whether property has a garage

Part 1 — Data preprocessing

Apply the following steps to preprocess the data before estimation:

1. Recode the string values in column CentralAir into numbers such that 'N' is mapped to 0 and 'Y' is mapped to 1.
2. Recode the string values in column BuildingType and create the new variable IsSingleFamily, which takes on the value 1 whenever a house is a single-family home and 0 otherwise.
3. Convert the variables SalePrice, LivingArea, and LotArea to (natural) logs. Name the transformed columns logSalePrice, logLivingArea, and logLotArea.
4. Plot the histograms of SalePrice, LivingArea, and LotArea. In a new figure, plot the histograms of logSalePrice, logLivingArea, and logLotArea. Which set of variables is better suited for model fitting?

Part 2 — Model features

Model specification

You are now asked to estimate the following model of house prices as a function of house characteristics:

$$\begin{aligned}\log(\text{SalePrice}_i) = & \mu + f\left(\log(\text{LivingArea}_i), \log(\text{LotArea}_i), \text{OverallQuality}_i, \right. \\ & \left. \text{Bathrooms}_i, \text{Bedrooms}_i\right) \\ & + \gamma_0 \text{YearBuilt}_i + \gamma_1 \text{CentralAir}_i + \gamma_2 \text{IsSingleFamily}_i + \epsilon_i\end{aligned}$$

where i indexes observations and ϵ_i is an additive error term. The function $f(\bullet)$ is a *polynomial of degree 3* in its arguments, i.e., it includes all terms and interactions of the given variables where the exponents sum to 3 or less:

$$\begin{aligned}f(\log(\text{LivingArea}_i), \log(\text{LotArea}_i), \dots) = & \beta_0 \log(\text{LivingArea}_i) + \beta_1 \log(\text{LivingArea}_i)^2 \\ & + \beta_2 \log(\text{LivingArea}_i)^3 + \beta_3 \log(\text{LotArea}_i) \\ & + \beta_4 \log(\text{LotArea}_i)^2 + \beta_5 \log(\text{LotArea}_i)^3 \\ & + \beta_6 \log(\text{LivingArea}_i) \log(\text{LotArea}_i) \\ & + \beta_7 \log(\text{LivingArea}_i)^2 \log(\text{LotArea}_i) \\ & + \beta_8 \log(\text{LivingArea}_i) \log(\text{LotArea}_i)^2 \\ & + \dots\end{aligned}$$

Creating model features and outcomes

1. Complete the template code below to create a feature matrix X that contains all polynomial interactions as well as the remaining non-interacted variables.

Hints:

- Use the `PolynomialFeatures` transformation to create the polynomial terms and interactions from the columns `logLivingArea`, `logLotArea`, `OverallQuality`, `Bathrooms`, and `Bedrooms`.
- Make sure that the generated polynomial does *not* contain a constant (“bias”). You should include the intercept when estimating a model instead.
- You can use `np.hstack()` to concatenate two matrices (the polynomials and the remaining covariates) along the column dimension.
- The complete feature matrix X should contain a total of 58 columns (55 polynomial interactions and 3 non-polynomial features).

2. Split the data into a training and a test subset such that the training sample contains 70% of the observations.

Hint:

- Use the function `train_test_split()` to split the sample. Pass the argument `random_state=1234` to get reproducible results.
- Make sure to define the training and test samples only *once* so that they are identical for all estimators used below.

```
[9]: # Random state (for train/test split and cross-validation)
RANDOM_STATE = 1234

# Name of target variable
target = 'logSalePrice'
```

```

# Features included as polynomials
features_poly = [
    'logLivingArea',
    'logLotArea',
    'OverallQuality',
    'Bathrooms',
    'Bedrooms',
]

# Other features not included in polynomials
features_other = ['YearBuilt', 'CentralAir', 'IsSingleFamily']
features = features_poly + features_other

# Keep only columns that are used to estimate model
columns = [target] + features
df = df[columns]

# Response variable
y = df[target]

# TODO: Create polynomial features

# TODO: Merge polynomial features and non-polynomial features into single matrix X

# TODO: Split data into training and test sets

```

Part 3 — Linear regression

Perform the following tasks:

1. Estimate the above specification using the linear regression model `LinearRegression` on the training subset.
 - Do you need to standardize features before estimating a linear regression model?
 - Does the linear regression model have any hyperparameters?
2. Compute and report the root mean squared error (RMSE) and the R^2 on the test sample.

Hints:

- The root mean squared error can be computed with `root_mean_squared_error()`.
- The R^2 can be computed with `r2_score()`.

Part 4 — Ridge regression

Next, you want to estimate a Ridge regression model, which has the regularization strength α as a hyperparameter.

1. Use the template code below to run `RidgeCV` to determine the best α on the training subsample. You can use the MSE metric (the default) to find the optimal α . Report the optimal α and the corresponding MSE.
 - Does Ridge regression require feature standardization? If so, do not forget to apply it before fitting the model.
2. Use the function `plot_validation_curve()` defined below to plot the MSE (averaged over folds on the training subsample) against the regularization strength α .
3. Compute and report the RMSE and the R^2 on the test sample.

Hints: - Create RidgeCV with store_cv_results=True to store the MSEs on all folds. - The MSEs for all folds and alphas are stored in the attribute cv_results_ after fitting. - The (negative!) best MSE is stored in the attribute best_score_ after fitting.

```
[14]: import matplotlib.pyplot as plt

def plot_validation_curve(alphas, mse_mean, title=None):
    """
    Plot validation curve for Ridge or Lasso.

    Parameters
    -----
    alphas : array-like
        Regularization strengths.
    mse_mean : array-like
        Cross-validated MSE (averaged over folds).
    title : str, optional
        Title of the plot.
    """

    # Index of MSE-minimizing alpha
    imin = np.argmin(mse_mean)

    # Plot MSE against alphas, highlight minimum MSE
    plt.plot(alphas, mse_mean)
    plt.xlabel(r'Regularization strength $\alpha$ (log scale)')
    plt.ylabel('Cross-validated MSE')
    plt.scatter(alphas[imin], mse_mean[imin], s=15, c='black', zorder=100)
    plt.axvline(alphas[imin], ls=':', lw=0.75, c='black')
    plt.title(title)
    plt.xscale('log')
```

```
[15]: # TODO: Manually transform features

# TODO: Create alpha grid uniformly spaced in logs on [1e-6, 100]
# alphas =

# TODO: Create RidgeCV and fit model
# ridge_cv =

# TODO: Report the best alpha and the corresponding MSE score

# TODO: Compute MSEs averaged across folds (stored in cv_results_)
# mse_mean =

# TODO: Plot validation curve
# plot_validation_curve(alphas, mse_mean)

# TODO: Compute and report RMSE and R2 on the test sample
```

Part 5 — Lasso

Next, you want to estimate a Lasso model, which also has a regularization strength hyperparameter α :

1. Use the template below to run `LassoCV` to determine the best α on the training subsample using cross-validation with 5 folds. You can use the MSE metric (the default) to find the optimal α . Report the optimal α and the corresponding MSE.
 - Does Lasso require feature standardization? If so, do not forget to apply it before fitting the model.

2. Use the function `plot_validation_curve()` to plot the MSE (averaged over folds on the training subsample) against the regularization strength α .
3. Compute and report the RMSE and the R^2 on the test sample for the model using the optimal α .
4. Report the number of nonzero coefficients for the model using the optimal α .

Hints:

- Getting Lasso to converge may require some experimentation. The following settings should help:
 1. Increase the maximum number of iterations to `max_iter=100_000`.
 2. Use `selection='random'` and set `random_state=1234` to get reproducible results.
- Use `eps=1.0e-4` as an argument to `LassoCV` to specify the ratio of the smallest to the largest α .
- After cross-validation is complete, the MSE for each value of α and each fold is stored in the attribute `mse_path_`, which is an array with shape `(N_ALPHA, N_FOLDS)`.

```
[20]: # TODO: Run cross-validation using LassoCV. Use the transformed features from earlier.
# lasso_cv =

# TODO: Report the best alpha (stored in alpha_)

# TODO: Compute MSEs averaged across folds (stored in mse_path_)
# mse_mean =

# TODO: Plot validation curve
# plot_validation_curve(lasso_cv.alphas_, mse_mean)

# TODO: Compute and report RMSE and R2 on the test sample

# TODO: Report number of nonzero coefficients (stored in coef_)
```

Part 6 — Compare estimation results

Create a table that contains the RMSE and R^2 computed on the test sample for all three models (using their optimal hyperparameters). Which model performs best?